Victor Nin

ECE 4425: Data Communications Lab

victor_nin13@gwu.edu

GNS3 Custom Subnets

Objective:

During this lab session the students were given the task to get more familiar with the GNS3 software with its features and tools. The main objective was to create the network diagram provided of four separate subnets which involved three PCs, two routers, and six switches. After that task was complete, the next step was to follow the directions and configure what is asked in the respective consoles and answer the questions in the lab instructions.

Background:

Once the student was able to download the GNS3 software, the goal was to run a simulation to see the IP addresses of multiple devices being pinged onto the network. This software allows users to emulate, configure, test, and troubleshoot virtual and real networks. It also allows one to run a topology using a number of devices on one's computer using multiple servers or even hosted in the cloud.

Experimental Setup:

In order to start the experiment, students must have to download all the required software in order to run the GNS3, such as VMware in order to get going. Once students finished this process, in GNS3, the task was to create the network diagram provided which involved three PCs, two routers, and six switches. This will allow us to create four separate subnets. After that task was complete, the next step was to follow the directions and configure what is asked in the respective consoles.

Simulation:

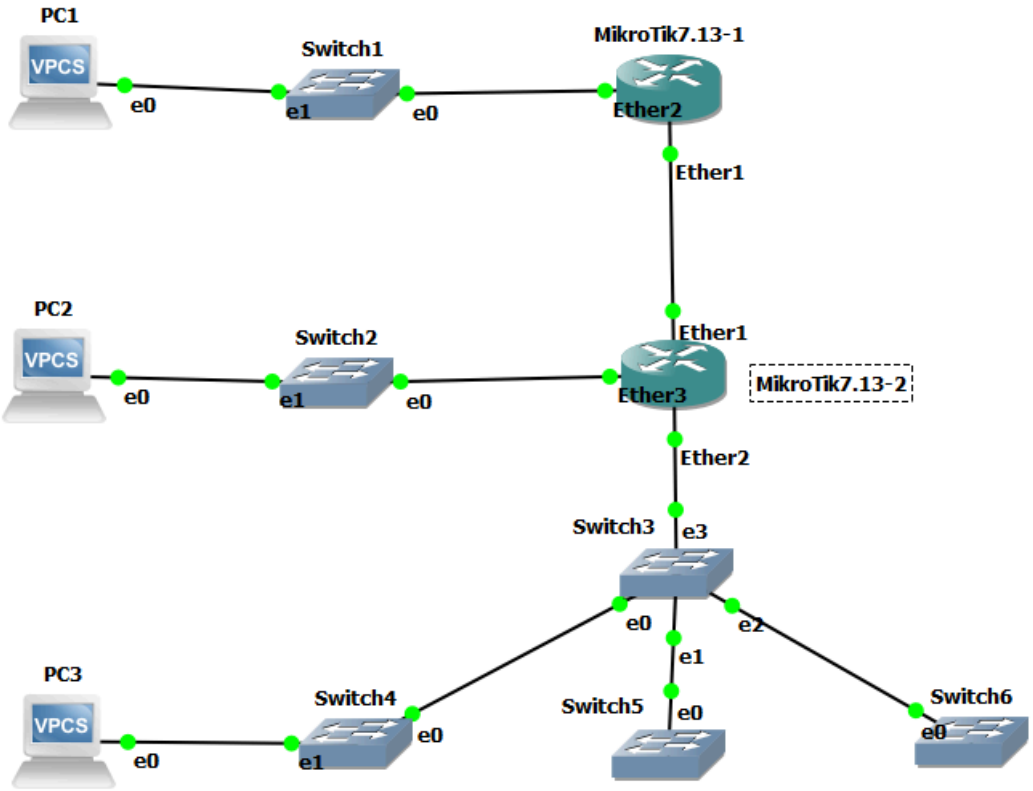


Figure 1 - Network Topology

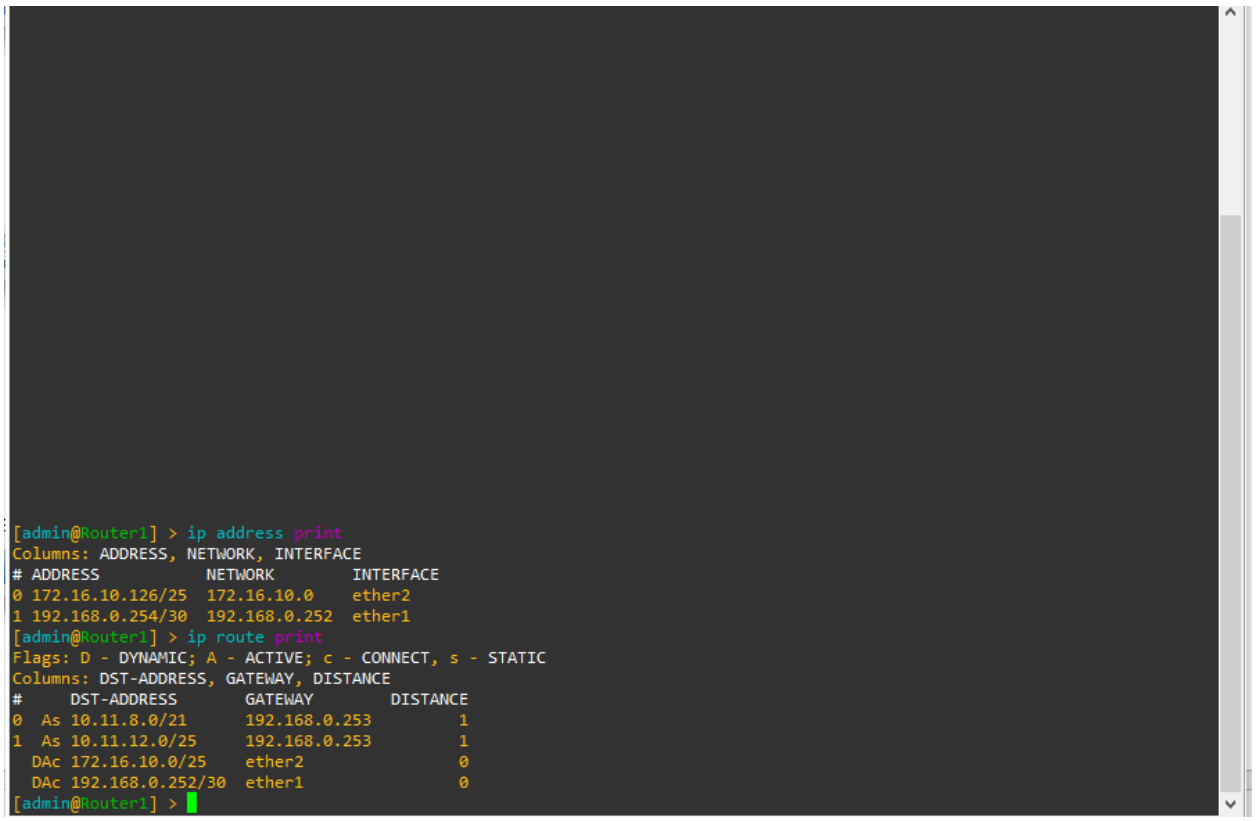


Figure 2 - Router 1 MicroTik Information

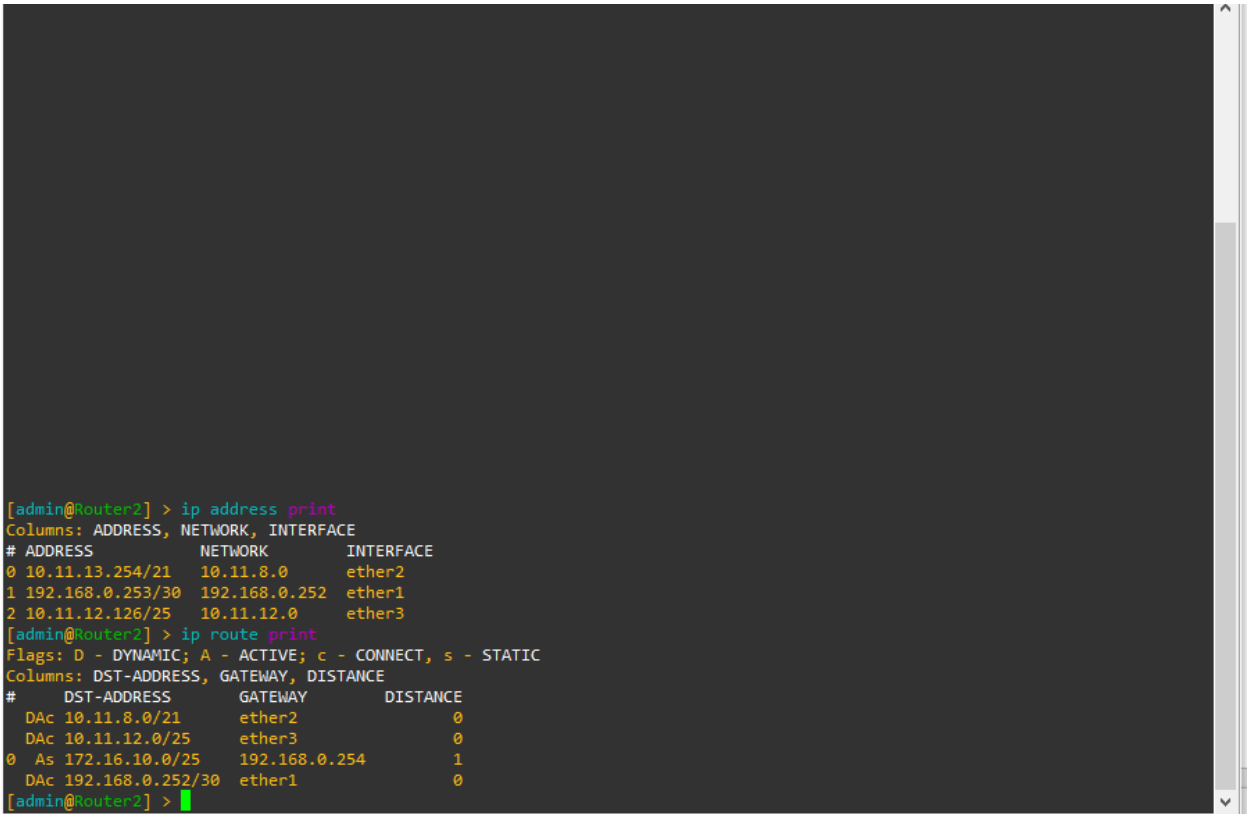


Figure 3 - Router 2 MicroTik Information

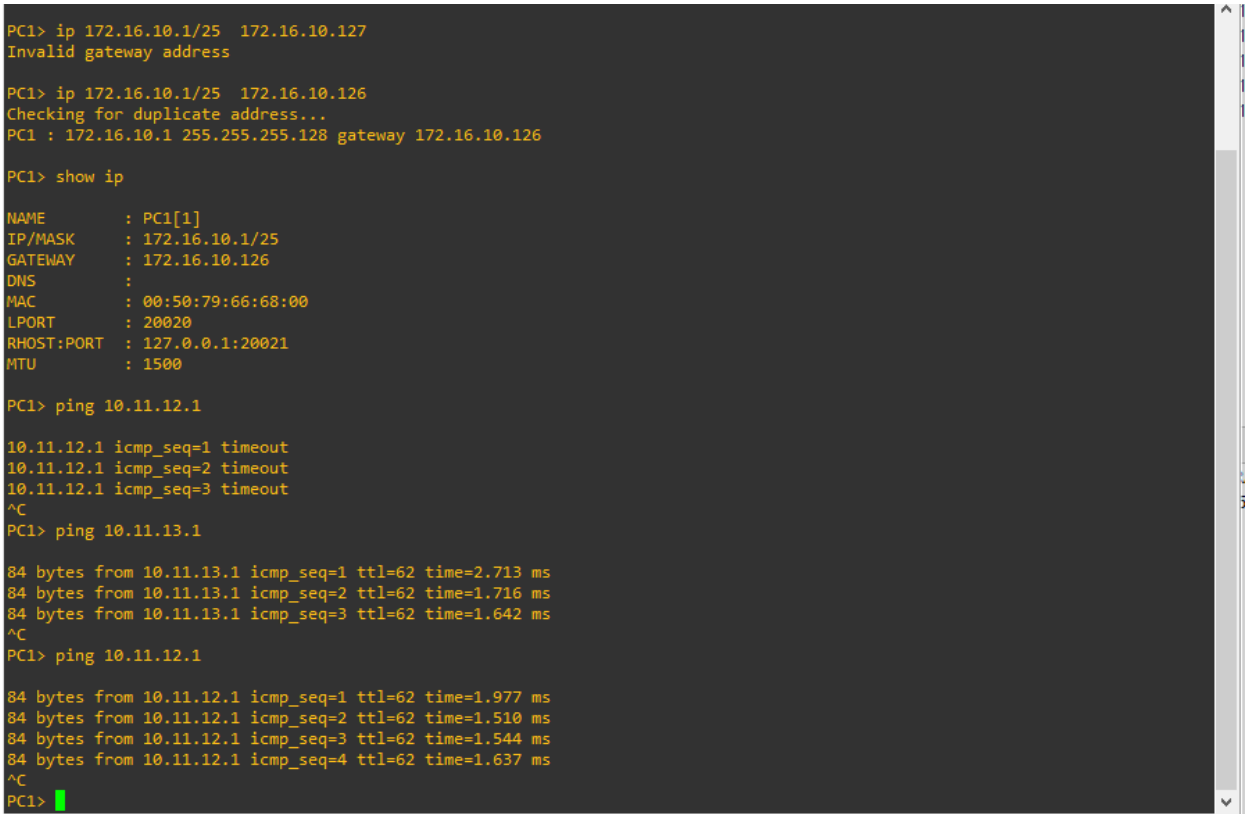


Figure 4 - Pings from PC1 to PC2 & PC3



Figure 5 - PC2 Pings

Subnet #	Subnet Address a.b.c.d/n	Netmask	Number of IPs in subnet	Lowest Usable Host Address	Highest Usable Host Address
1	172.16.10.0/25	11111111.11111111.11111111.10000000	128	172.16.10.1	172.16.10.126
2	10.11.12.0/25	11111111.11111111.11111111.10000000	128	10.11.12.1	10.11.12.126
3	10.11.13.0/21	11111111.11111111.11111000.00000000	2048	10.11.8.0	10.11.15.254
4	192.168.0.254/30	11111111.11111111.11111111.11111100	4	192.168.0.253	192.168.0.254

Table 1- Subnets

Host	IP Address a.b.c.d/n	Subnet Address a.b.c.d/n
PC1	172.16.10.1/25	172.16.10.0/25
PC2	10.11.12.1/25	10.11.12.0/25
PC3	10.11.13.1/21	10.11.13.0/21
Router1, Ether1 Interface	192.168.0.254/30	192.168.0.0/30
Router1, Ether2 Interface	172.16.10.126/25	172.16.10.0/25
Router2, Ether1 Interface	192.168.0.253/30	192.168.0.0/30
Router2, Ether2 Interface	10.11.13.254/21	10.11.13.0/21

Router2, Ether3 Interface	**10.11.12.126/25**	**10.11.12.0/25**

Table 2 - IP Address for all Host Interfaces

Analysis:

In this lab session, we got to understand and work with configuring multiple PCs and routers using subnets and routing tables. Specifically, we learned how to configure how to route unreachable PCs by ip routing them in the router's configuration. This would then allow us to ping these PCs in different subnets to each other in the respective PC's configuration. A subnet as we learned is class is a part of a network or a network inside a network. The router determines whether a IP address is in a different subnet based on the IP addresses. In the lab, we now had to utilize the using different kinds of hosts for certain networks and making sure we used the correct ranges for host addresses. The important information provided in **Table 1** and **Table 2** were a key component in organizing our network, subnets, hosts, and routing in order to get our desired results.

Conclusion:

In conclusion, the lab provided students with access to the GNS3 softwares along with added experience to its tools and commands. The objective of testing the PC connections with our routers to understand what was going on, and what information was being requested and sent proved to be very informative. The most important aspect was understanding how to configure the router and add a routing table in order to connect distant subnets. The most important part of this lab in order to complete our objective was to make sure that we used the correct host addresses based on the

network we were using. This caused some problems in this lab but once it was fixed everything worked smoothly.